

Methodology

Literature Search

Relevant research papers were collected using PubMed. The search focused on studies involving Large Language Models (LLMs), antibiotic prescribing, antimicrobial stewardship, and antimicrobial resistance.

Screening Process

The review was completed in two stages:

1. Title and abstract screening
2. Papers not directly related to the topic were removed.
3. Full-text review
4. Remaining papers were read in detail and relevant information was extracted.

Data Extraction

For each selected study, the following information was recorded:

- Title
- Authors
- Publication year
- Journal
- Methodology
- Models evaluated
- Key findings
- Limitations
- Future research directions
- Policy and ethical considerations

Analysis

The collected studies were compared to identify common trends, strengths, and limitations of LLMs in healthcare applications related to antibiotic prescribing and antimicrobial resistance.

The findings were also reviewed in the context of existing AI governance frameworks, including guidance from the World Health Organization (WHO) and the Organisation for Economic Co-operation and Development (OECD).

Outcome

The review provides an overview of the current research landscape and highlights areas where additional validation, transparency, and governance measures may be required before LLMs can be safely integrated into clinical practice.